

PROTEIN PRACTICAL QUESTION

1. The term "protein" is derived from the Greek word proteios, which means:

- A. Building block
- B. First place
- C. Life-giver
- D. Complex chain

Answer: B. First place. The Greek word proteios means "of first importance" or "first place," reflecting the essential role of proteins in life.

2. Which scientist suggested the name "proteins" for the group of organic compounds utmost important to life?

- A. Mendel
- B. Berzelius
- C. Watson
- D. Pasteur

Answer: B. Berzelius. The Swedish chemist Jöns Jacob Berzelius suggested the name "proteins" in 1838.

3. Proteins are described as the most abundant and functionally diverse molecules in:

- A. The atmosphere
- B. Living systems
- C. The Earth's crust
- D. Non-organic matter

Answer: B. Living systems. Proteins constitute about 50% of the dry weight of cells and perform a vast array of functions in living organisms.

4. All proteins share the common structural feature of being:

- A. Branched polymers of sugars
- B. Linear polymers of amino acids
- C. Circular chains of lipids
- D. Random clusters of nucleotides

Answer: B. Linear polymers of amino acids. Proteins are unbranched polymers formed from amino acids linked by peptide bonds.

5. How many different L- α -amino acids are commonly found as constituents of mammalian proteins?

- A. 100
- B. 300
- C. 20
- D. 27,000

Answer: C. 20. There are 20 standard L- α -amino acids that are commonly found in mammalian proteins.

6. The specific amino acid sequence of a protein is determined by:

- A. Dietary intake
- B. The genetic code in DNA
- C. The temperature of the cell
- D. Random chance

Answer: B. The genetic code in DNA. The amino acid sequence of a protein is encoded by the nucleotide sequence of genes in DNA.

7. At physiologic pH (~ 7.4), the carboxyl group of a free amino acid is:

- A. Protonated ($-\text{COOH}$)
- B. Dissociated ($-\text{COO}^-$)
- C. Attached to a lipid
- D. Completely neutral

Answer: B. Dissociated ($-\text{COO}^-$). At physiologic pH, the pK_a of the carboxyl group (~ 2.2) is much lower than 7.4, so it exists predominantly in its deprotonated (dissociated) form.

8. Free amino acids in solution at physiologic pH exist as:

- A. Anions
- B. Cations
- C. Zwitterions
- D. Non-polar molecules

Answer: C. Zwitterions. At physiologic pH, amino acids exist as dipolar ions (zwitterions) with both a positive charge on the amino group and a negative charge on the carboxyl group.

9. The length of polypeptide chains in proteins typically ranges from:

- A. 10–50 amino acids
- B. 100–1,800 amino acids
- C. 5,000–10,000 amino acids
- D. Exactly 20 amino acids

Answer: B. 100–1,800 amino acids. Most polypeptide chains in proteins range from 100 to 1,800 amino acid residues.

10. The largest known protein in the human body, found in muscle, is:

- A. Hemoglobin
- B. Cytochrome c
- C. Titin
- D. Insulin

Answer: C. Titin. Titin is the largest known protein, found in skeletal and cardiac muscle, with a molecular weight of approximately 3,000 kDa.

11. The linear sequence of amino acids in a polypeptide chain is called the:

- A. Primary structure
- B. Secondary structure
- C. Tertiary structure
- D. Quaternary structure

Answer: A. Primary structure. The primary structure refers to the linear sequence of amino acids in a polypeptide chain.

12. By convention, the amino acid sequence of a protein is read from the:

- A. C-terminal to N-terminal
- B. Methyl end to Carboxyl end
- C. N-terminal to C-terminal
- D. 3' end to 5' end

Answer: C. N-terminal to C-terminal. The convention is to write and read amino acid sequences from the free amino (N-terminal) end to the free carboxyl (C-terminal) end.

13. Amino acids in proteins are joined covalently by:

- A. Hydrogen bonds
- B. Ionic bonds

- C. Peptide bonds
- D. Van der Waals forces

Answer: C. Peptide bonds. Amino acids are covalently linked by peptide bonds, which are formed between the carboxyl group of one amino acid and the amino group of another.

14. A peptide bond is chemically defined as an:

- A. Ester linkage
- B. Amide linkage
- C. Ether linkage
- D. Glycosidic bond

Answer: B. Amide linkage. A peptide bond is a type of amide linkage formed between the carboxyl group of one amino acid and the amino group of another.

15. Which characteristic describes the peptide bond?

- A. It is easily broken by heating
- B. It has partial double-bond character and is planar
- C. It is highly flexible and rotates 360 degrees
- D. It is ionized at physiologic pH

Answer: B. It has partial double-bond character and is planar. The peptide bond has partial double-bond character due to resonance, making it rigid and planar.

16. Secondary structure consists of local regions of polypeptide chains stabilized by:

- A. Disulfide bridges
- B. Repeating patterns of hydrogen bonds
- C. Hydrophobic interactions
- D. Covalent links between side chains

Answer: B. Repeating patterns of hydrogen bonds. Secondary structures like α -helices and β -sheets are stabilized by repeating patterns of hydrogen bonds between backbone atoms.

17. Which is the most common spiral secondary structure in proteins?

- A. β -sheet
- B. β -bend
- C. α -helix
- D. Random coil

Answer: C. α -helix. The α -helix is the most common spiral secondary structure in proteins.

18. Which amino acid acts as a "helix breaker" and is not usually found in α -helices?

- A. Glycine
- B. Alanine
- C. Proline
- D. Leucine

Answer: C. Proline. Proline acts as a helix breaker because its rigid ring structure prevents the proper rotation needed for α -helix formation.

19. β -Bends (reverse turns) are generally composed of how many amino acids?

- A. Two
- B. Four
- C. Six
- D. Ten

Answer: B. Four. β -Bends (reverse turns) typically consist of four amino acid residues.

20. The overall three-dimensional folding of a single polypeptide chain is the:

- A. Primary structure
- B. Secondary structure
- C. Tertiary structure
- D. Quaternary structure

Answer: C. Tertiary structure. The tertiary structure refers to the overall three-dimensional folding of a single polypeptide chain.

21. In globular proteins, the tertiary structure generally forms a core that is:

- A. Densely packed and hydrophobic
- B. Hollow and water-filled
- C. Rich in charged amino acids
- D. Composed only of glycine

Answer: A. Densely packed and hydrophobic. The core of globular proteins is densely packed with hydrophobic amino acid side chains.

22. Physically independent regions of a large protein that fold independently are called:

- A. Motifs
- B. Domains
- C. Protomers

D. Zymogens

Answer: B. Domains. Domains are physically independent regions of a large protein that fold into stable structures independently.

23. The arrangement of multiple polypeptide subunits in a protein is called:

- A. Primary structure
- B. Supersecondary structure
- C. Tertiary structure
- D. Quaternary structure

Answer: D. Quaternary structure. The quaternary structure refers to the arrangement of multiple polypeptide subunits in a protein.

24. In hemoglobin, the unit structure composed of one α and one β chain is called a:

- A. Monomer
- B. Protomer
- C. Domain
- D. Motif

Answer: B. Protomer. The $\alpha\beta$ dimer in hemoglobin is called a protomer.

25. Which proteins are required for the proper folding of many species of proteins in the cell?

- A. Enzymes
- B. Hormones
- C. Chaperones (Heat-shock proteins)
- D. Antibodies

Answer: C. Chaperones (Heat-shock proteins). Chaperones assist in the proper folding of many proteins in the cell.

26. Proteins composed exclusively of amino acid residues are classified as:

- A. Conjugated proteins
- B. Simple proteins
- C. Complex proteins
- D. Metalloproteins

Answer: B. Simple proteins. Simple proteins are composed exclusively of amino acid residues.

27. Conjugated proteins contain a non-protein component called a:

- A. Holoprotein
- B. Side chain
- C. Prosthetic group
- D. Subunit

Answer: C. Prosthetic group. Conjugated proteins contain a non-protein component called a prosthetic group.

28. Proteins that resemble irregular balls and are usually soluble in aqueous media are:

- A. Fibrous proteins
- B. Globular proteins
- C. Transmembrane proteins
- D. Scleroproteins

Answer: B. Globular proteins. Globular proteins resemble irregular balls and are usually soluble in aqueous media.

29. Fibrous proteins are characterized as being:

- A. Highly soluble in water
- B. Geometrically linear and structural
- C. Compactly folded into spheres
- D. Primarily involved in catalysis

Answer: B. Geometrically linear and structural. Fibrous proteins are linear, structural proteins that provide support and shape.

30. Which of the following is a classic example of a fibrous protein?

- A. Myoglobin
- B. Hemoglobin
- C. Collagen
- D. Albumin

Answer: C. Collagen. Collagen is a classic example of a fibrous protein.

31. Proteins conjugated with lipids are called:

- A. Glycoproteins
- B. Phosphoproteins
- C. Lipoproteins

D. Metalloproteins

Answer: C. Lipoproteins. Proteins conjugated with lipids are called lipoproteins.

32. A protein that contains a carbohydrate group of less than 4% is typically a:

A. Mucoprotein

B. Glycoprotein

C. Lipoprotein

D. Phosphoprotein

Answer: B. Glycoprotein. Glycoproteins typically contain less than 4% carbohydrate.

33. Metalloproteins conjugated with iron include:

A. Ceruloplasmin

B. Carbonic anhydrase

C. Hemoglobin and Myoglobin

D. Casein

Answer: C. Hemoglobin and Myoglobin. These are iron-containing metalloproteins.

34. Which plasma protein maintains the majority of oncotic pressure in the blood?

A. Globulin

B. Fibrinogen

C. Albumin

D. Transferrin

Answer: C. Albumin. Albumin maintains approximately 80% of plasma oncotic pressure.

35. Albumins are characterized by being:

A. Insoluble in water

B. Readily soluble in water and coagulated by heat

C. Only soluble in dilute salt solutions

D. Part of the cell membrane only

Answer: B. Readily soluble in water and coagulated by heat. Albumins are soluble in water and coagulated by heat.

36. Proteins that act as catalysts to increase the rate of biochemical reactions are:

A. Hormones

B. Antibodies

- C. Enzymes
- D. Receptors

Answer: C. Enzymes. Enzymes are proteins that act as biological catalysts.

37. Muscle contraction is performed by which mechanical proteins?

- A. Albumin and Globulin
- B. Actin and Myosin
- C. Collagen and Elastin
- D. Keratin and Fibrin

Answer: B. Actin and Myosin. These two proteins are responsible for muscle contraction.

38. Which protein is responsible for transporting oxygen in the blood?

- A. Myoglobin
- B. Hemoglobin
- C. Transferrin
- D. Ferritin

Answer: B. Hemoglobin. Hemoglobin transports oxygen in the blood from the lungs to the tissues.

39. Transferrin is a β -globulin synthesized in the liver that binds and transports:

- A. Copper
- B. Oxygen
- C. Iron
- D. Glucose

Answer: C. Iron. Transferrin transports iron in the blood.

40. Which protein serves as the primary cellular storage site for iron?

- A. Hemoglobin
- B. Transferrin
- C. Ferritin
- D. Albumin

Answer: C. Ferritin. Ferritin is the primary cellular storage protein for iron.

41. Immunoglobulins (antibodies) are produced by B-cells to provide:

- A. Structural support
- B. Catalysis

- C. Protection against infection
- D. Energy storage

Answer: C. Protection against infection. Antibodies are produced by B-cells to protect against infections.

42. Collagen acts like "steel cables" in reinforced concrete to provide a framework for:

- A. Blood clotting
- B. Bone and connective tissue
- C. Nerve impulses
- D. Muscle storage

Answer: B. Bone and connective tissue. Collagen provides structural framework for bone and connective tissues.

43. Keratin is a major structural component of:

- A. The liver and spleen
- B. Hair, nails, and skin
- C. The inner mitochondrial membrane
- D. Red blood cells

Answer: B. Hair, nails, and skin. Keratin is the major structural protein in hair, nails, and skin.

44. α 1-Antitrypsin is a serine protease inhibitor that prevents the degradation of:

- A. Collagen
- B. Elastin in the lungs
- C. Keratin in the skin
- D. Myosin in muscle

Answer: B. Elastin in the lungs. α 1-Antitrypsin inhibits proteases that degrade elastin in the lung tissue.

45. Deficiency of α 1-antitrypsin, particularly in smokers, results in:

- A. Scurvy
- B. Emphysema
- C. Edema
- D. Osteogenesis imperfecta

Answer: B. Emphysema. Deficiency of α 1-antitrypsin leads to progressive lung damage and

emphysema, especially in smokers.

46. Altered versions of normal proteins that cause neurodegenerative diseases (like Mad Cow Disease) are:

- A. Chaperones
- B. Prions
- C. Zymogens
- D. Isoforms

Answer: B. Prions. Prions are misfolded proteins that cause neurodegenerative diseases.

47. Alzheimer's disease is characterized by the formation of neurotoxic assemblies called:

- A. Hemoglobin polymers
- B. Amyloid protein
- C. Keratin filaments
- D. Collagen fibrils

Answer: B. Amyloid protein. Alzheimer's disease is characterized by the accumulation of amyloid protein plaques.

48. Posttranslational modification of collagen requires which cofactor for the hydroxylation of proline?

- A. Vitamin B12
- B. Vitamin D
- C. Vitamin C (ascorbate)
- D. Iron

Answer: C. Vitamin C (ascorbate). Vitamin C is required as a cofactor for proline hydroxylation in collagen synthesis.

49. Scurvy is a disease caused by a deficiency of Vitamin C, leading to:

- A. Excessive hair growth
- B. Unstable collagen and fragile blood vessels
- C. High blood sugar
- D. Excessive muscle mass

Answer: B. Unstable collagen and fragile blood vessels. Vitamin C deficiency leads to defective collagen synthesis, causing fragile blood vessels and connective tissue.

50. The addition of phosphate groups to serine, threonine, or tyrosine residues is called:

- A. Glycosylation
- B. Carboxylation
- C. Phosphorylation
- D. Acetylation

Answer: C. Phosphorylation. Phosphorylation is the addition of phosphate groups to serine, threonine, or tyrosine residues.

51. Glycosylation refers to the addition of:

- A. Lipids
- B. Carbohydrates (sugars)
- C. Metals
- D. Phosphates

Answer: B. Carbohydrates (sugars). Glycosylation is the addition of carbohydrate groups to proteins.

52. The addition of γ -carboxyglutamate to clotting factors requires:

- A. Vitamin C
- B. Vitamin K
- C. Vitamin A
- D. Biotin

Answer: B. Vitamin K. Vitamin K is required for the γ -carboxylation of glutamate residues in clotting factors.

53. The "central dogma" of molecular biology describes the flow of information from:

- A. Protein to DNA
- B. DNA to RNA to Protein
- C. RNA to Lipid
- D. Sugar to ATP

Answer: B. DNA to RNA to Protein. The central dogma describes the flow of genetic information from DNA to RNA to protein.

54. Proteins that perform the same function but have different primary structures are:

- A. Protomers
- B. Isoforms (Isozymes)

- C. Zymogens
- D. Chaperones

Answer: B. Isoforms (Isozymes). Isoforms are proteins that perform the same function but have different primary structures.

55. Which enzyme catalyzes the transfer of phosphate from ATP to creatine?

- A. Alanine aminotransferase
- B. Creatine kinase (CK)
- C. Lactate dehydrogenase
- D. Amylase

Answer: B. Creatine kinase (CK). Creatine kinase catalyzes the transfer of phosphate from ATP to creatine.

56. The CK-MB isoenzyme is primarily diagnostic for damage to the:

- A. Brain
- B. Liver
- C. Heart (Myocardial infarction)
- D. Skeletal muscle

Answer: C. Heart (Myocardial infarction). CK-MB is elevated in myocardial infarction.

57. Protein denaturation results in the loss of:

- A. Primary structure only
- B. Secondary and tertiary structure
- C. The genetic code
- D. Individual amino acids

Answer: B. Secondary and tertiary structure. Denaturation disrupts secondary and tertiary structures while preserving primary structure.

58. Which reagent is used for the sequential sequencing of a peptide from the N-terminal?

- A. Ninhydrin
- B. Edman reagent (phenylisothiocyanate)
- C. Urea
- D. Cyanogen bromide

Answer: B. Edman reagent (phenylisothiocyanate). Edman degradation is used for sequential sequencing from the N-terminus.

59. What is the approximate total concentration of plasma proteins in the blood?

- A. 1–5 g/L
- B. 10–20 g/L
- C. 65–80 g/L
- D. 500 g/L

Answer: C. 65–80 g/L. Normal plasma protein concentration ranges from 65–80 g/L.

60. A decrease in serum albumin levels below 20 g/L often leads to:

- A. Dehydration
- B. Edema (water entry into IF)
- C. High blood pressure
- D. Increased clotting

Answer: B. Edema (water entry into IF). Low albumin reduces plasma oncotic pressure, leading to edema.

61. Haptoglobin (α 2-globulin) is essential for conserving iron by binding to:

- A. Free oxygen
- B. Free hemoglobin
- C. Glucose
- D. Cholesterol

Answer: B. Free hemoglobin. Haptoglobin binds free hemoglobin to conserve iron.

62. The "antioxidant" role of Transferrin comes from preventing the formation of:

- A. ATP
- B. Hydroxyl radicals (via the Fenton reaction)
- C. Carbon dioxide
- D. Peptide bonds

Answer: B. Hydroxyl radicals (via the Fenton reaction). Transferrin prevents iron from catalyzing hydroxyl radical formation.

63. Which protein is rich in basic amino acids like Histidine and Lysine and associates with DNA?

- A. Albumin
- B. Histones (Globins)
- C. Collagen

D. Elastin

Answer: B. Histones (Globins). Histones are rich in basic amino acids and associate with DNA.

64. α -Keratin rigidity is determined by the number of:

- A. Hydrogen bonds
- B. Ionic bonds
- C. Disulfide bonds
- D. Peptide bonds

Answer: C. Disulfide bonds. The rigidity of α -keratin is determined by the number of disulfide bonds.

65. Collagen Type I is most prevalent in which tissues?

- A. Cartilage
- B. Tendons and Bone
- C. Basement membranes
- D. Blood vessels

Answer: B. Tendons and Bone. Type I collagen is the major collagen in bone, tendons, and skin.

66. Type IV collagen is classified as:

- A. Fibril-forming
- B. Network-forming
- C. Fibril-associated
- D. Liquid-based

Answer: B. Network-forming. Type IV collagen forms a network structure and is found in basement membranes.

67. The precursor for elastin is a linear polypeptide called:

- A. Procollagen
- B. Tropoelastin
- C. Fibrillin
- D. Desmosine

Answer: B. Tropoelastin. Tropoelastin is the precursor of elastin.

68. Desmosine cross-links in elastin are formed from which amino acid?

- A. Glycine
- B. Proline
- C. Lysine
- D. Valine

Answer: C. Lysine. Desmosine cross-links are formed from lysine residues in elastin.

69. Myoglobin consists of how many polypeptide chains?

- A. One
- B. Two
- C. Four
- D. Eight

Answer: A. One. Myoglobin is a single polypeptide chain protein.

70. Hemoglobin A consists of:

- A. One α and one β chain
- B. Two α and two β chains
- C. Four α chains
- D. Three α chains

Answer: B. Two α and two β chains. Hemoglobin A is a tetramer with two α and two β chains.

71. Which molecule binds reversibly to the heme group in Hb and Mb?

- A. Carbon dioxide
- B. Glucose
- C. Oxygen
- D. Nitrogen

Answer: C. Oxygen. Oxygen binds reversibly to the heme group in hemoglobin and myoglobin.

72. The Bohr effect refers to the release of oxygen from hemoglobin in response to:

- A. High oxygen levels
- B. Lower pH (increased H^+) and increased CO_2
- C. High altitudes
- D. Lower temperature

Answer: B. Lower pH (increased H^+) and increased CO_2 . The Bohr effect describes

increased oxygen release at lower pH and higher CO₂.

73. Glycosylated hemoglobin (HbA1c) is used to monitor:

- A. Liver function
- B. Long-term blood glucose control in diabetes
- C. Oxygen saturation
- D. Muscle damage

Answer: B. Long-term blood glucose control in diabetes. HbA1c reflects average blood glucose levels over 2–3 months.

74. Nonenzymatic glycosylation of proteins can lead to:

- A. Better protein function
- B. Advanced glycosylation end products (AGEs)
- C. Faster protein degradation
- D. Increased ATP production

Answer: B. Advanced glycosylation end products (AGEs). Nonenzymatic glycosylation produces AGEs, which can damage proteins.

75. Which is the most abundant protein in the human body?

- A. Albumin
- B. Hemoglobin
- C. Collagen
- D. Myosin

Answer: C. Collagen. Collagen is the most abundant protein in the human body, constituting about 25–35% of total body protein.

76. In collagen, which amino acid is found at every third position?

- A. Proline
- B. Glycine
- C. Lysine
- D. Alanine

Answer: B. Glycine. Glycine is found at every third position in the collagen triple helix (Gly-X-Y repeat).

77. Proteins in the plasma move in an electric field based on their:

- A. Size only

- B. Color
- C. Net charge
- D. Age

Answer: C. Net charge. Plasma proteins migrate in electrophoresis based on their net charge.

78. Which protein is found in fireflies and reacts with ATP to produce light?

- A. GFP
- B. Hemoglobin
- C. Luciferin
- D. Casein

Answer: C. Luciferin. Luciferin is the protein in fireflies that reacts with ATP to produce light.

79. Green Fluorescent Protein (GFP) was originally isolated from:

- A. Fireflies
- B. Jellyfish (*Aequorea victoria*)
- C. Snakes
- D. Rhinoceros horn

Answer: B. Jellyfish (*Aequorea victoria*). GFP was originally isolated from the jellyfish *Aequorea victoria*.

80. Snake venoms and plant toxins like ricin are examples of:

- A. Storage proteins
- B. Exploitive/Toxic proteins
- C. Clotting factors
- D. Hormones

Answer: B. Exploitive/Toxic proteins. Venoms and plant toxins are examples of toxic proteins.

81. Which protein transports 90% of serum copper?

- A. Albumin
- B. Transferrin
- C. Ceruloplasmin
- D. Hemoglobin

Answer: C. Ceruloplasmin. Ceruloplasmin transports approximately 90% of serum copper.

82. Carbonic anhydrase is a metalloprotein that contains:

- A. Iron
- B. Copper
- C. Zinc
- D. Magnesium

Answer: C. Zinc. Carbonic anhydrase contains zinc as a cofactor.

83. Phosphoproteins like casein of milk have phosphate attached to:

- A. Hydroxyl groups of Serine or Threonine
- B. Carboxyl groups of Glutamate
- C. Amino groups of Lysine
- D. Methyl groups of Alanine

Answer: A. Hydroxyl groups of Serine or Threonine. Phosphate is attached to the hydroxyl groups of serine or threonine in phosphoproteins.

84. Histones form ionic bonds with DNA because they are:

- A. Negatively charged
- B. Highly hydrophobic
- C. Positively charged
- D. Made of lipids

Answer: C. Positively charged. Histones are positively charged, allowing them to form ionic bonds with negatively charged DNA.

85. Scleroproteins (animal skeleton proteins) include all of the following EXCEPT:

- A. Keratin
- B. Collagen
- C. Elastin
- D. Insulin

Answer: D. Insulin. Insulin is a globular protein, not a scleroprotein.

86. Gliadins are proteins found in:

- A. Eggs
- B. Milk

C. Cereals (Gluten)

D. Blood

Answer: C. Cereals (Gluten). Gliadins are gluten proteins found in wheat and other cereals.

87. The primary standard for evaluating dietary protein quality is:

A. Caloric density

B. PDCAAS (Protein Digestibility-Corrected Amino Acid Score)

C. Total weight

D. Sweetness

Answer: B. PDCAAS (Protein Digestibility-Corrected Amino Acid Score). PDCAAS is the standard method for evaluating protein quality.

88. How many amino acids are "essential" (cannot be synthesized by humans)?

A. 5

B. 9

C. 20

D. 300

Answer: B. 9. There are 9 essential amino acids that cannot be synthesized by humans.

89. A clinical symptom of protein-energy malnutrition (Kwashiorkor) is:

A. Hyperactivity

B. Edema

C. Increased height

D. Strengthened bones

Answer: B. Edema. Edema is a hallmark sign of Kwashiorkor due to low albumin levels.

90. In the "Mediterranean diet," monounsaturated fats and proteins from fish are favored over:

A. Vegetables

B. Saturated fats from red meat

C. Fruits

D. Water

Answer: B. Saturated fats from red meat. The Mediterranean diet favors fish and monounsaturated fats over saturated fats from red meat.

91. What is the function of "antifreeze protein" in certain fish?

- A. Increases blood pressure
- B. Prevents ice crystal formation in tissues
- C. Stores fat
- D. Catalyzes glycolysis

Answer: B. Prevents ice crystal formation in tissues. Antifreeze proteins prevent ice crystal formation in fish living in freezing waters.

92. Transmembrane proteins often contain one or more regions that are:

- A. Random coils
- B. Membrane-spanning α -helices
- C. Soluble in water
- D. Composed of pure sugar

Answer: B. Membrane-spanning α -helices. Transmembrane proteins often have hydrophobic α -helical regions that span the membrane.

93. Which proteins use ATP energy to transport cargo toward the "plus" end of microtubules?

- A. Dyneins
- B. Kinesins
- C. Actins
- D. Myosins

Answer: B. Kinesins. Kinesins transport cargo toward the plus end of microtubules using ATP energy.

94. The hydrolysis and resynthesis of 300–400 g of body protein each day is called:

- A. Denaturation
- B. Translation
- C. Protein turnover
- D. Glycosylation

Answer: C. Protein turnover. Protein turnover is the continuous breakdown and resynthesis of body proteins.

95. Proteins marked for rapid degradation in the cytosol are often tagged with:

- A. Phosphate
- B. Ubiquitin
- C. Glucose

D. Heme

Answer: B. Ubiquitin. Ubiquitin is a tag that marks proteins for degradation.

96. The cytosolic complex that degrades ubiquitinated proteins is the:

A. Mitochondrion

B. Proteasome

C. Ribosome

D. Golgi

Answer: B. Proteasome. The proteasome is responsible for degrading ubiquitinated proteins.

97. Which vitamin is essential for blood clotting protein carboxylation?

A. Vitamin C

B. Vitamin K

C. Vitamin E

D. Vitamin D

Answer: B. Vitamin K. Vitamin K is essential for γ -carboxylation of blood clotting proteins.

98. The primary function of plasma lipoproteins is to:

A. Catalog DNA

B. Transport hydrophobic lipids in aqueous plasma

C. Act as enzymes for digestion

D. Provide structural support for hair

Answer: B. Transport hydrophobic lipids in aqueous plasma. Lipoproteins transport hydrophobic lipids in the blood.

99. What happens to the oxygen affinity of hemoglobin as the pH decreases?

A. It increases

B. It decreases (O₂ is released)

C. It stays the same

D. It becomes zero

Answer: B. It decreases (O₂ is released). Decreasing pH reduces hemoglobin's oxygen affinity (Bohr effect).

100. The unique three-dimensional shape of a functional protein is its:

- A. Denatured state
- B. Native conformation
- C. Primary sequence
- D. Random coil

Answer: B. Native conformation. The native conformation is the functional three-dimensional shape of a protein.

